# RUOSHI XU

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## RESEARCH INTERESTS

Multimodal Intelligence, Medical AI, Agentic Systems

## EDUCATION&EXPERIENCE

**Southern University of Science and Technology** **Sept. 2022 - Jun. 2026**
**B.Eng.** in Intelligence Science and Technology (**Turing Class**) **GPA: 3.90 / 4.00**, **Rank: 1/23**
Supervised by Prof. Jianguo Zhang

**Hengqin Laboratory** **Jul. 2026 - Present**
**Research Intern (Paid)** Supervised by Prof. Shengda Luo

## PUBLICATIONS & RESEARCH

**CIRSeg: Coarse-to-Fine Intensity-Robust Liver Segmentation with Source-Free Continual Test-Time Adaptation** [Paper]
**Ruoshi Xu**, Mingqi Gao, Shengda Luo, Jingkun Chen.
*MICCAI 2026 CARE Workshop.*
• Led the project end-to-end, including the design and implementation of the segmentation framework, all experiments, manuscript writing, and poster preparation.
• Achieved 97.13%/97.93% Dice and 20.18/11.30 mm HD95 on the ID/OOD test sets (official results), respectively; ranked **2nd among 112 teams on the GED4 validation set**.

**ACME: Annotation-Compatible Model-Driven Exploration for Single-Source Domain Generalization in Medical Image Segmentation** [Project]
Shengda Luo, **Ruoshi Xu***, Jingkun Chen.
*Submitted to IEEE Transactions on Medical Imaging (TMI).*
• Designed and implemented the complete Probe–Qualify–Learn framework; led all experimental design and evaluation, authored the Method and Experiments sections.
• Outperformed the best compared methods across four benchmarks, improving Dice by 4.44 pp (Fundus), 2.79/0.13 pp (MSCMR, bidirectional), 7.50 pp (Polyp), and 10.27 pp (Prostate).

**WorldGenBench: A World-Knowledge-Integrated Benchmark for Reasoning-Driven Text-to-Image Generation** [Paper]
**Ruoshi Xu**, Daoan Zhang*, ···, Shengda Luo.
*arXiv:2505.01490.*
• Contributed to model evaluation using benchmark metrics and research visualizations; led subsequent manuscript revisions, rebuttal preparation, and resubmission efforts.
• Revealed substantial gaps in world-knowledge implicit reasoning among text-to-image models, with GPT-4o achieving the highest Knowledge Checklist Scores of 24.46 (Humanities) and 19.61 (Nature).

**Reinforcing 3D Understanding in Point-VLMs via Geometric Reward Credit Assignment** [Paper]
Jingkun Chen, **Ruoshi Xu***, Mingqi Gao, Shengda Luo, Jungong Han.
*Submitted to IEEE Transactions on Multimedia (TMM).*
• Proposed geometric reward credit assignment, designed and implemented the ShapeNetCore-based synthetic data generation pipeline, and conducted all training and experimental evaluations.
• Improved 3D Key Point Accuracy from 0.64 to 0.93, achieving 0.686 in 3D bounding box IoU and 0.852 in reprojection consistency while preserving robust 2D localization performance.